

Gebze Technical University

Department of Mechanical Engineering

ME 331 System Dynamics and Vibration

Undamped Free Vibration

Single-degree-of-freedom systems

Saeed Lotfan

Today's topics

1. Modeling: Newton, D'Alembert and virtual work
2. Natural frequency and static equilibrium
3. Initial conditions, amplitude and phase
4. Energy exchange and conservation
5. Torsional vibration
6. Rayleigh: spring mass, U-tube and beam models
7. Assumptions and result checks

From a structure to a vibration model



Choose a coordinate.

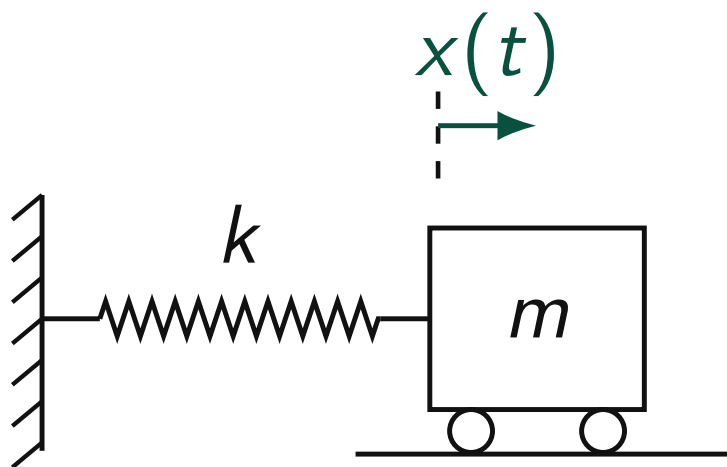
Identify inertia and restoring action.

State the idealization.

3.1 Modeling

Undamped spring–mass model

One coordinate ■ Linear spring ■ Frictionless motion

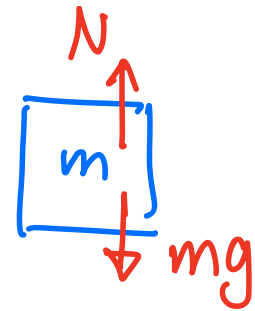
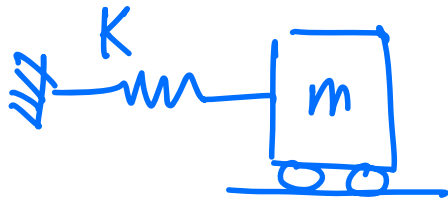


$x = 0$: equilibrium

$$m\ddot{x} + kx = 0$$

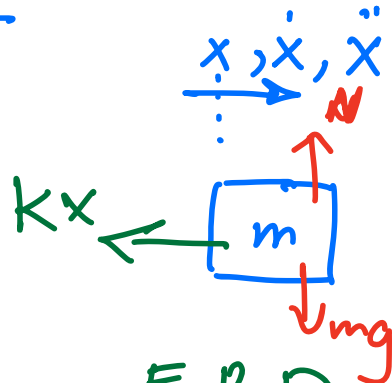
Displacement from
equilibrium

Whiteboard notes



F.B.D

B.M



F.B.D

A.M

$$+ \rightarrow \sum F_{\text{FBD.AM}} \equiv \sum F_{\text{K.I.D}} \\ -Kx = m\ddot{x}$$

K.D

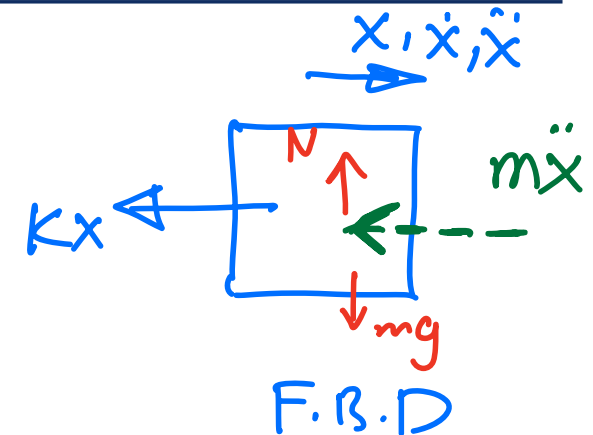
$$m \ddot{x}(t) + Kx(t) = 0$$

Equation of motion

3.1 Modeling

D'Alembert and virtual displacement

Newton's law as dynamic equilibrium



$$\sum F_x - m\ddot{x} = 0$$

$$\sum F_x = 0$$

A compatible imagined displacement at a fixed instant

$$\delta W = (-kx - m\ddot{x}) \delta x = 0$$

$$-kx - m\ddot{x} = 0$$

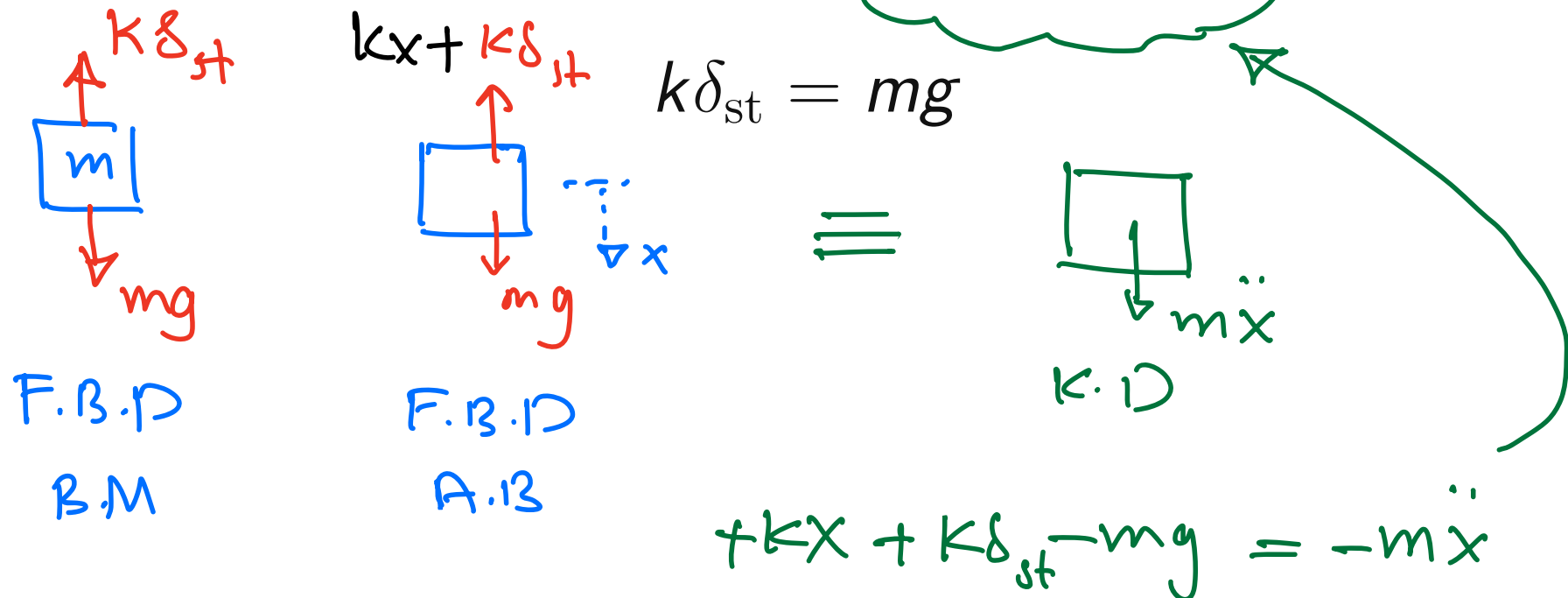
$$m\ddot{x} + kx = 0$$

Why is δx arbitrary? Which reactions do no work?

3.1 Modeling

Vertical spring-mass system

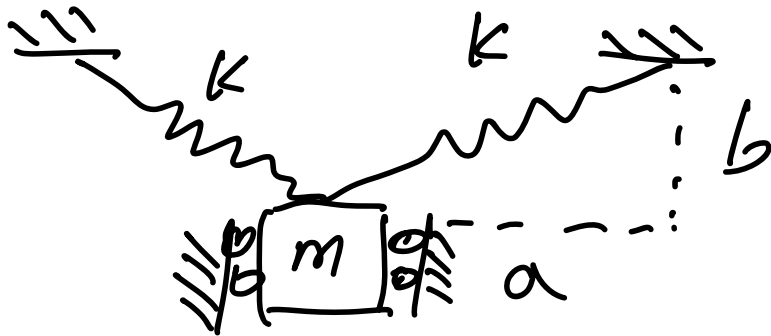
Measure x from static equilibrium.



Whiteboard notes

in vertical $m-k \rightarrow$ we ignore mg !

Can we do it always for any $m-k$!



3.2 Natural frequency

Natural frequency and period

$$m\ddot{x}(t) + Kx(t) = 0$$

$$\ddot{x}(t) + \omega_n^2 x(t) = 0$$

Natural angular frequency

Frequency and period

$$\omega_n = \sqrt{\frac{k}{m}}$$

rad/s

$$f_n = \frac{\omega_n}{2\pi} \quad \text{Hz}$$

$$T_n = \frac{2\pi}{\omega_n} \quad \text{s} = \frac{1}{f_n}$$

How do mass and stiffness change the motion?

3.2 Natural frequency

Frequency from static deflection

Static test ■ Units ■ Parameter effects

$$\omega_n = \sqrt{\frac{k}{m}}, \quad k\delta_{st} = mg$$



Whiteboard notes

$$m\ddot{x} + Kx = 0 \rightarrow x(t) = x_h + \cancel{x_p}^{\rightarrow 0}$$

$$x(t) = Ce^{st} \rightarrow \dot{x}(t) = Cs e^{st}, \ddot{x} = Cs^2 e^{st}$$

$$mCs^2 e^{st} + KCe^{st} = 0$$

$$Ce^{st} (ms^2 + K) = 0 \quad ms^2 + K = 0 \quad s = \pm i \sqrt{\frac{K}{m}}$$

Whiteboard notes

$$m\ddot{x} + kx = 0 \quad \xrightarrow{\text{has solution}} \quad x(t) = Ce^{st}$$

$$s = \pm i\sqrt{\frac{k}{m}} \rightarrow x(t) = C_1 \underbrace{e^{i\sqrt{\frac{k}{m}}t}} + C_2 e^{-i\sqrt{\frac{k}{m}}t}$$

$$e^{i\alpha} = \cos \alpha + i \sin \alpha$$

$$C_1 \left(\cos \sqrt{\frac{k}{m}}t + i \sin \sqrt{\frac{k}{m}}t \right) + C_2 \left(\cos \sqrt{\frac{k}{m}}t - i \sin \sqrt{\frac{k}{m}}t \right)$$

$$\downarrow$$
$$x(t) = A \underbrace{\sin \sqrt{\frac{k}{m}}t}_{\text{a harmonic}} + B \cos \sqrt{\frac{k}{m}}t$$

Whiteboard notes

$$X(t) = A \sin \omega_n t + B \cos \omega_n t$$

$$\omega_n = \sqrt{\frac{k}{m}}$$

$$\begin{cases} m\ddot{x} + kx = 0 \\ x(0) = x_0 \end{cases}$$

$$\dot{x}(0) = v_0$$

3.3 Initial conditions and response

Response to initial conditions

Two initial conditions determine the motion.

$$x(0) = x_0, \quad \dot{x}(0) = v_0$$

$$x(t) = A_1 \cos(\omega_n t) + A_2 \sin(\omega_n t)$$

3.3 Initial conditions and response

$$m\ddot{x} + Kx = 0 \quad 1 \text{ DoF.}$$

Apply the initial conditions

Find A_1, A_2 ■ Write $x(t)$ ■ Check $t = 0$

$$x(t) = A_1 \cos \omega_n t + A_2 \sin \omega_n t$$

$$x(0) = A_1 \quad \leftarrow \quad x(0) = A_1 (\underbrace{1}) + A_2 (0)$$

$$\dot{x}(0) = A_2 \omega_n \quad \leftarrow \quad \dot{x}(0) = -A_1 \omega_n \underbrace{\sin \omega_n t}_0 + A_2 \omega_n \underbrace{\cos \omega_n t}_1$$

Whiteboard notes

$$X(0) = A_1 = x_0 \quad \dot{X}(0) = A_2 \omega_n = v_0$$

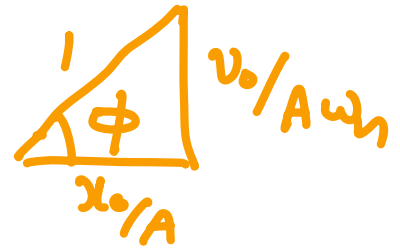
$$X(t) = A_1 \cos \omega_n t + A_2 \sin \omega_n t$$

$$x(t) = \underbrace{x_0 \cos \omega_n t}_{\text{harmonic}} + \frac{v_0}{\omega_n} \underbrace{\sin \omega_n t}_{\text{harmonic}}$$

Solution of undamped free vibration
of 1DOF system

Whiteboard notes

$$x(t) = x_0 \cos \omega_n t + \frac{v_0}{\omega_n} \sin \omega_n t$$



$$A = \sqrt{x_0^2 + \frac{v_0^2}{\omega_n^2}} \xrightarrow{\text{Week 2:}}$$

$$A \cdot \left[\underbrace{\frac{x_0}{A}} \cos \omega_n t + \frac{v_0/\omega_n}{A} \sin \omega_n t \right]$$

$$= A \cos(\omega_n t - \phi)$$

$$\phi = \tan^{-1} \frac{v_0}{x_0 \omega_n}$$

3.3 Initial conditions and response

Amplitude and phase

Use the book's cosine form and check the initial conditions.

$$x(t) = A \cos(\omega_n t - \phi)$$

$$A = \sqrt{x_0^2 + \left(\frac{v_0}{\omega_n}\right)^2}$$

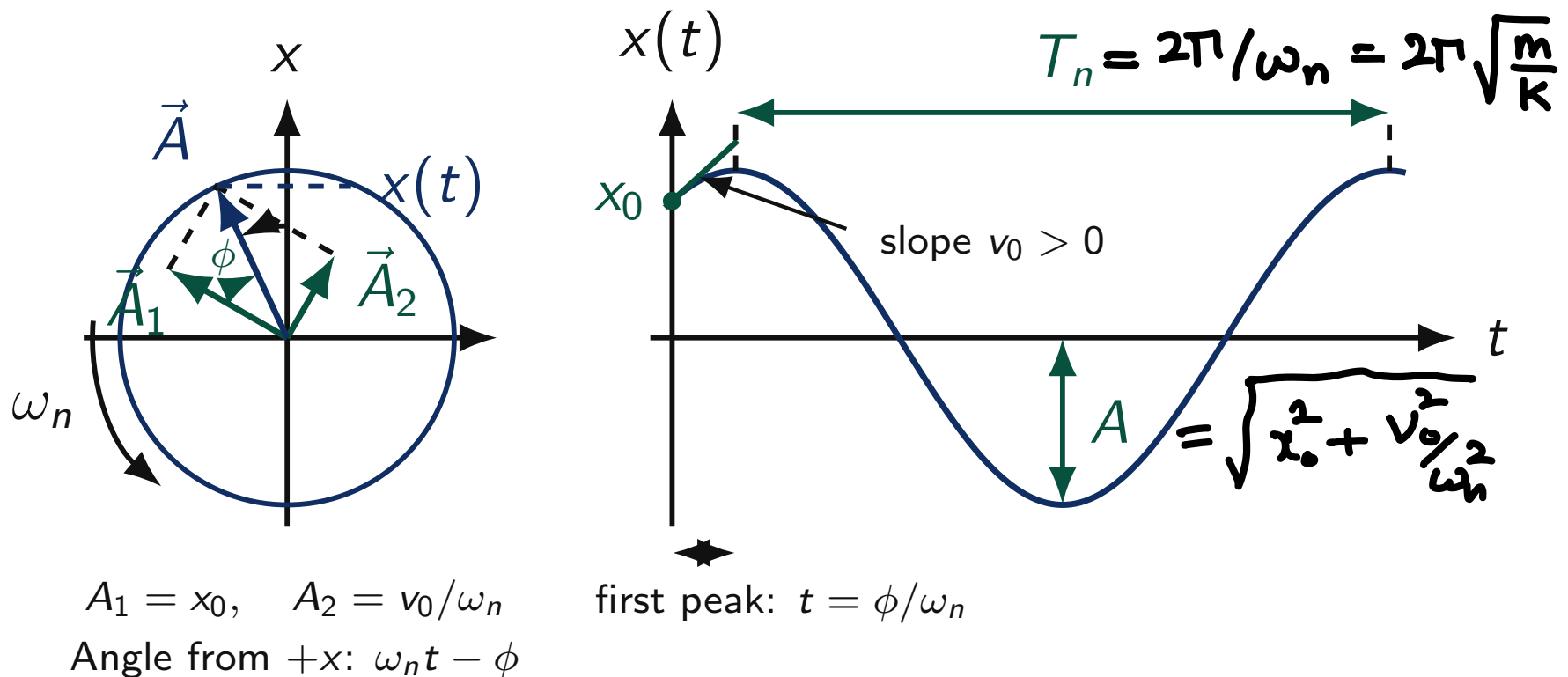
$$\tan \phi = \frac{v_0}{\omega_n x_0}$$

Choose the quadrant from x_0 and v_0 .

3.3 Initial conditions and response

Rotating amplitude vectors

Vertical projection gives $x(t) = A \cos(\omega_n t - \phi)$.



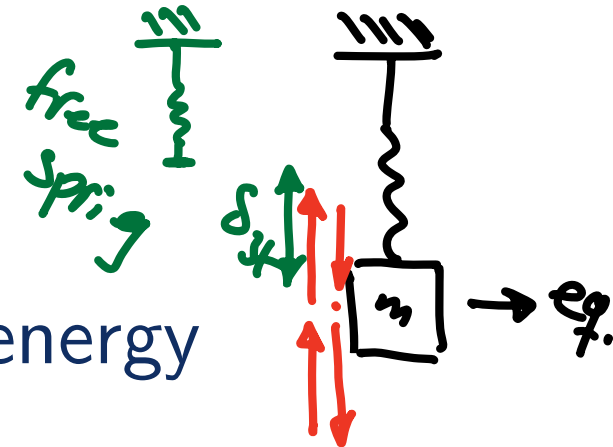
3.4 Energy

Energy exchange

Energy measured relative to static equilibrium

Kinetic energy

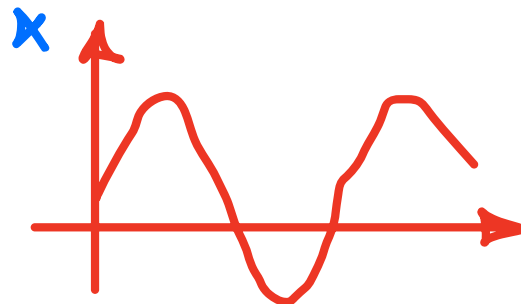
Potential energy



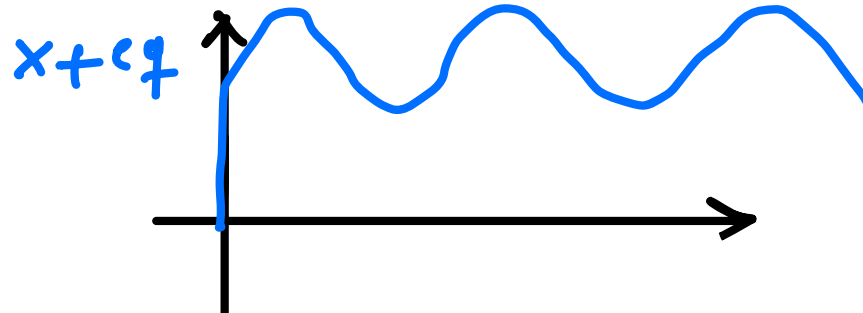
$$T = \frac{1}{2} m \dot{x}^2$$

$$U = \frac{1}{2} k x^2$$

At maximum displacement?



At equilibrium?



3.4 Energy

Conservation of energy

Compare two instants of the same motion.

$$\dot{E} = 0$$

$$E = T + U$$
$$E(t) = \frac{1}{2} m \dot{x}(t)^2 + \frac{1}{2} k x(t)^2$$

$$A = \sqrt{x_0^2 + \frac{v_0^2}{\omega_n^2}}$$

$$A^2 = x_0^2 + \frac{v_0^2}{\omega_n^2}$$

$$t = t_0 \quad x_0, v_0$$

$$E_0 = \frac{1}{2} m v_0^2 + \frac{1}{2} k x_0^2$$

$$t = t_1 \quad x = A, v = 0$$

$$E = 0 + \frac{1}{2} k A^2$$

$$\rightarrow E = \frac{1}{2} k x_0^2 + \frac{1}{2} k v_0^2 \cdot \frac{m}{k}$$

the same

Whiteboard notes

$$m\ddot{x} + Kx = 0 \quad \times \dot{x} \rightarrow m\ddot{x}\dot{x} + Kx\dot{x} = 0$$

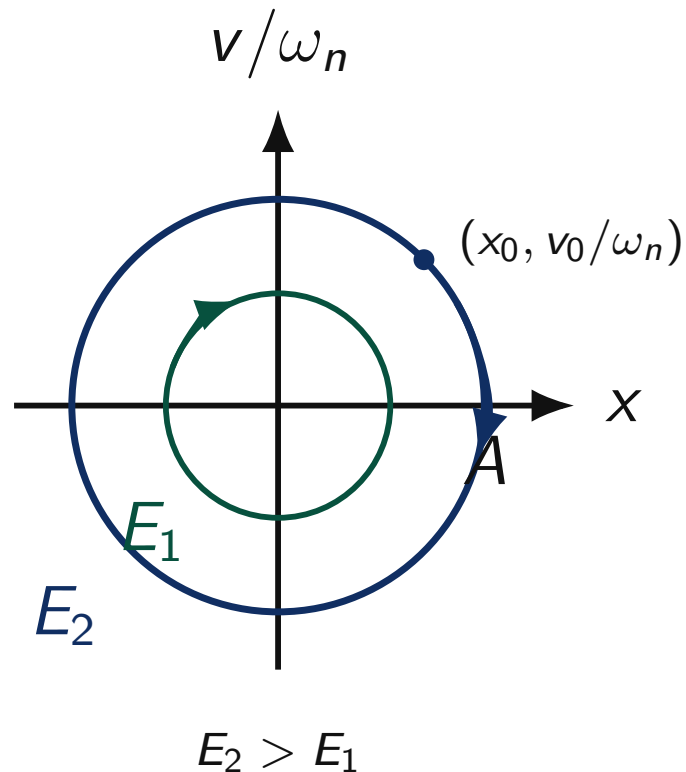
$$\int dt \Rightarrow \frac{1}{2} m \dot{x}^2 + \frac{1}{2} K x^2 = \text{const.}$$

3.4 Energy

$$X^2 + Y^2 = R^2$$

Phase plane and energy

Each orbit has constant $E = \frac{1}{2}kx^2 + \frac{1}{2}mv^2$.



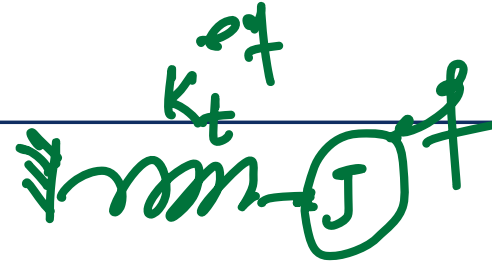
$$x^2 + \left(\frac{v}{\omega_n}\right)^2 = A^2$$

$$E = \frac{1}{2}kA^2$$

Clockwise motion

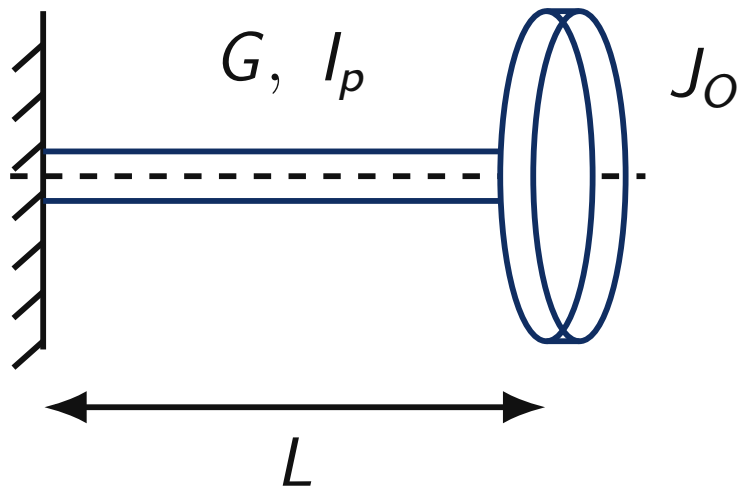
Larger orbit: more energy

3.5 Torsional vibration

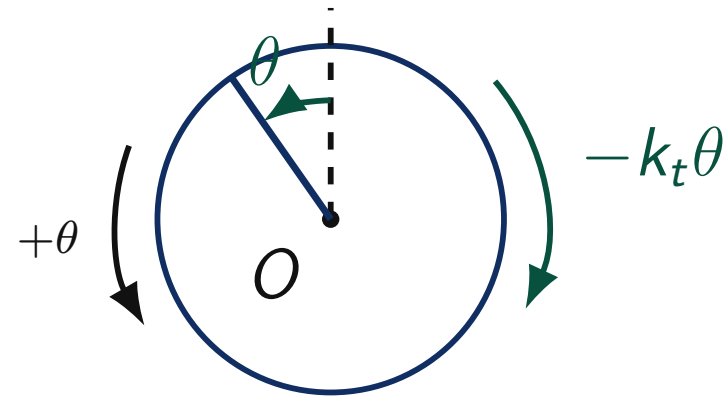


Undamped torsional vibration

Elastic shaft ■ Rigid disk ■ Fixed rotation axis



Shaft inertia neglected



End view: restoring torque

$$k_t = \frac{GI_p}{L}, \quad J_O \ddot{\theta} + k_t \theta = 0$$

3.5 Torsional vibration

Torsional response

Inertia axis ■ Restoring moment ■ Initial conditions

$$J_o \ddot{\theta} + K_t \theta = 0 \quad ; \quad \theta(0) = \theta_o, \quad \dot{\theta}(0) = \omega_o$$

$$\theta(t) = A \cos(\omega_n t - \Phi) \quad \omega_n = \sqrt{\frac{K_t}{J_o}}$$

$$A = \sqrt{\theta_o^2 + \frac{\omega_o^2}{\omega_n^2}}, \quad \Phi = \tan^{-1} \frac{\omega_o}{\theta_o \omega_n}$$

3.6 Rayleigh's method

Rayleigh energy method

Undamped motion about a stable equilibrium

$$x = A \cos \omega_n t$$

$$T_{\max} = \frac{1}{2} m \dot{A}^2 \omega_n^2$$

$$U_{\max} = \frac{1}{2} K A^2$$

$$T_{\max} = U_{\max}$$

Choose a coordinate. Express every motion using it.

$$\frac{1}{2} m \dot{A}^2 \omega_n^2 = \frac{1}{2} K A^2 \rightarrow \omega_n = \sqrt{\frac{K}{m}}$$

3.6 Rayleigh's method

Including the spring mass

Uniform spring mass m_s

▪

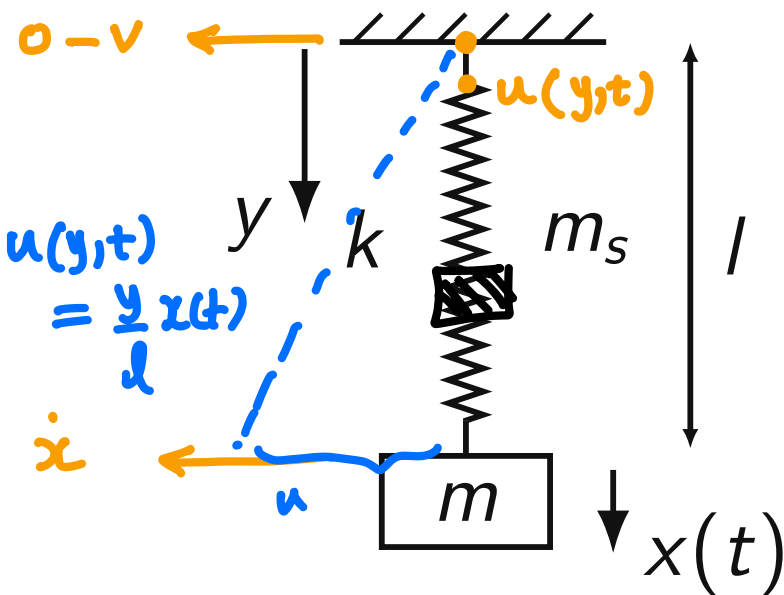
Fixed upper end

$$T = \frac{1}{2} m \dot{x}^2 + T_{\text{spring}}$$

$$T_{\text{spring}} = \frac{1}{2} \int_0^L \dot{u}^2 dm$$

Assume a linear displacement shape

$$u(y, t) = \frac{y}{l} x(t)$$



$$dm \quad \dot{u}(y, t)$$

$$dT = \frac{1}{2} dm \dot{u}^2$$

Whiteboard notes

$$T = \frac{1}{2} \int \dot{u}^2(y, t) dm$$

mass per length

$$dm = \rho dy$$

$$= \frac{1}{2} \int_0^l \rho \frac{y^2}{l^2} \dot{x}^2(t) dy$$

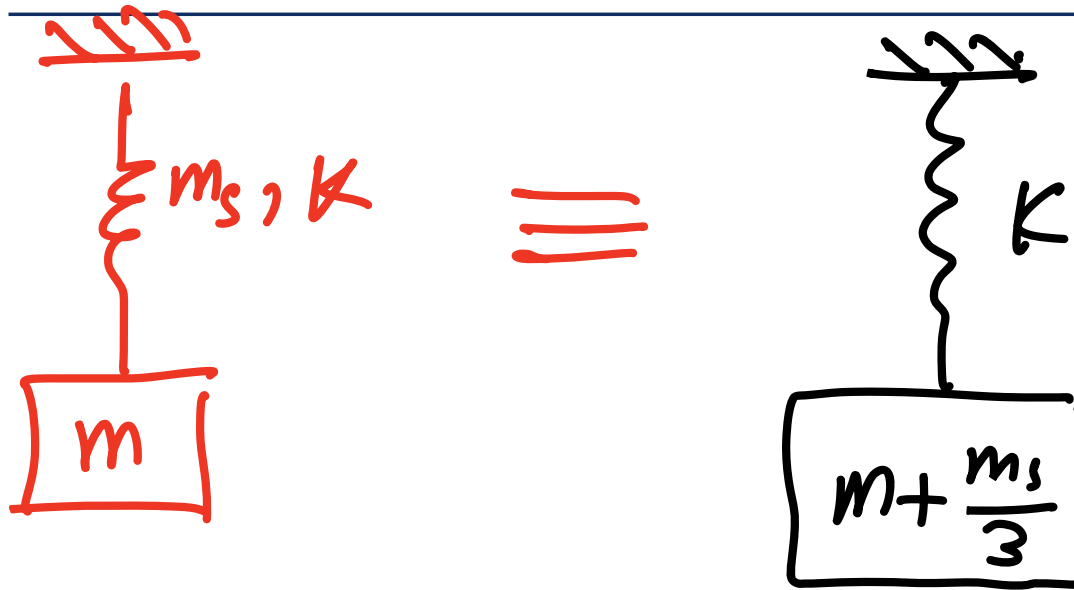
$$\dot{u}(y, t) = \frac{y}{l} \dot{x}(t)$$

$$= \frac{1}{2} \rho \frac{1}{l^2} \dot{x}^2(t) \underbrace{\int_0^l y^2 dy}_l = \frac{1}{2} \rho \frac{1}{l^2} \dot{x}^2 \left(\frac{l^3}{3} \right)$$

$$= \frac{1}{2} m_s \dot{x}^2 / 3$$

m_{eq}

Whiteboard notes



$$\omega_n = \sqrt{\frac{K}{m + m_s/3}}$$

Whiteboard notes

$$u(y,t) = ?$$

$$u(y,t) = \psi(y) \cdot \chi(t)$$

$$\psi(y) = \frac{y}{l} \quad \text{assumption}$$

$$u(y,t) = \frac{y}{l} \cdot \chi(t) \rightarrow \text{we solve just } \chi(t)$$

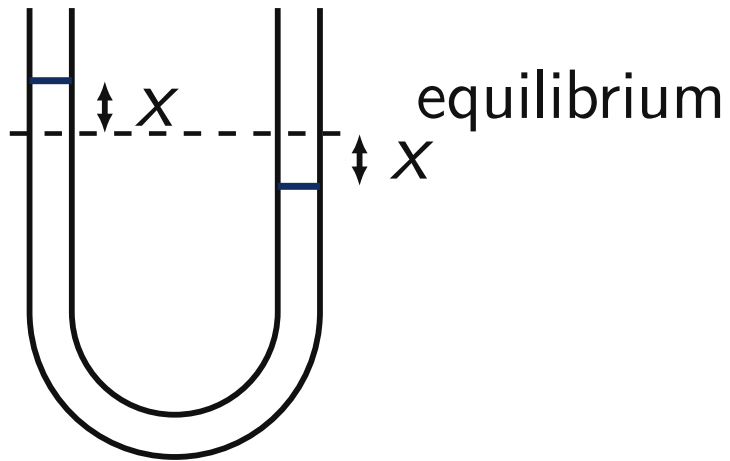
$$\psi(y) = \frac{y^2}{l^2} \quad ?$$

$$\psi(y) = \frac{y^2}{l^2} + 3\frac{y}{l}$$

3.6 Rayleigh's method

U-tube: natural frequency

Uniform area A ■ Liquid density ρ ■ Total length l



$$m_{\text{eq}} = \rho A l$$

$$k_{\text{eq}} = 2\rho g A$$

3.6 Rayleigh's method

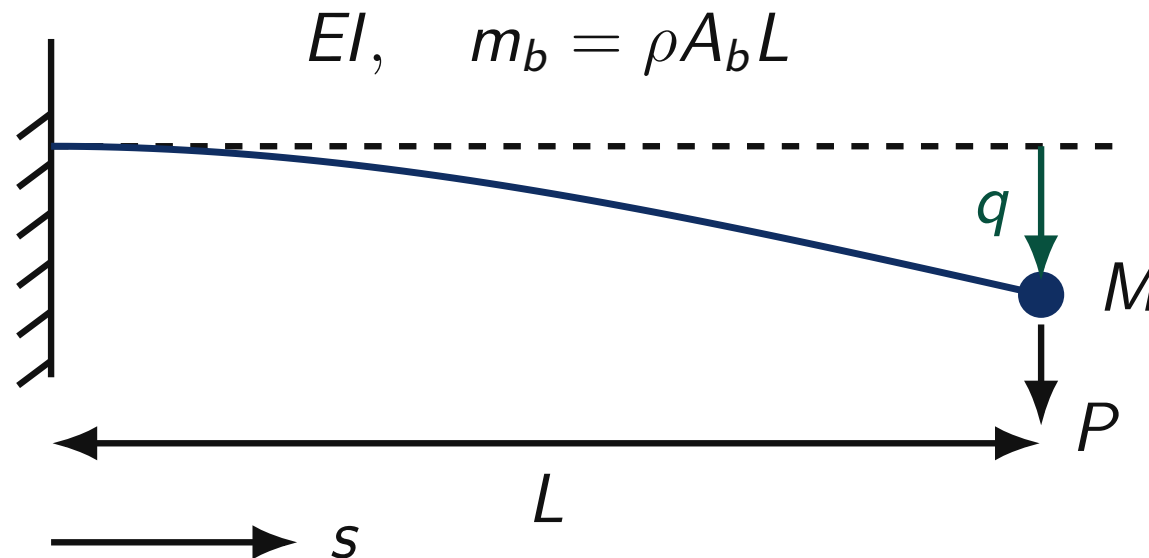
U-tube: energy calculation

Write $T_{\max}$ and $U_{\max}$, then find ω_n .

3.6 Rayleigh's method

Beam model: equivalent stiffness

Uniform cantilever ■ Tip displacement $q(t)$

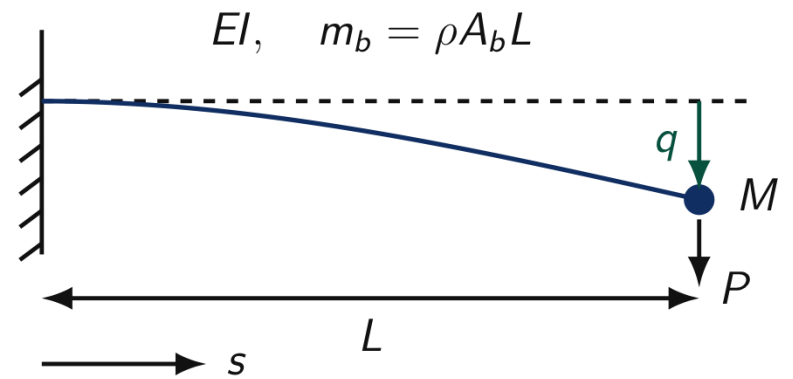


$$q_{\text{st}} = \frac{PL^3}{3EI} \quad \Rightarrow \quad k_{\text{eq}} = \frac{3EI}{L^3}$$

Whiteboard notes

If $M \gg m_b$

$$m_{eq} = M \quad K_{eq} = \frac{3EI}{L^3}$$



this was based on week 1+2 and static deflection!

$$M\ddot{q} + \frac{3EI}{L^3} q(t) = 0 \quad \text{Equation of motion!}$$

Whiteboard notes

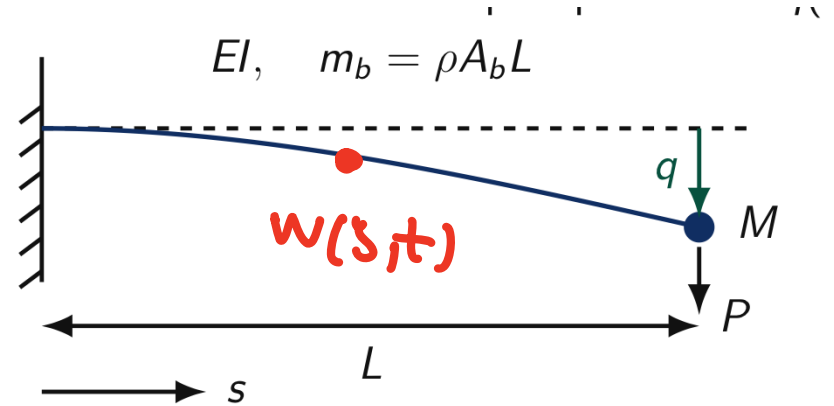
$$m_b \cancel{<} M$$

$$w(s,t) = \psi(s) q(t)$$

$$\psi(s) = ? \quad M(s) = P(L-s) \rightarrow \text{moment inside beam!}$$

static moment due to P !

Rayleighs method : $U = \int \sigma \epsilon dV$



Whiteboard notes

$$U = \int \sigma \epsilon dV = \int \frac{\sigma^2}{E} dV = \int E \epsilon^2 dV$$

$$\sigma = \frac{Mc}{I} \rightarrow U = \int \frac{AP^2(L-s)^2 c^2}{I^2 E} ds$$

$$M = P(L-s) \rightarrow EI w'' = P(L-s)$$

$$EI w' = P(Ls - \frac{s^2}{2}) + C_1$$

$$EI w = P(\frac{Ls^2}{2} - \frac{s^3}{6}) + C_1 s + C_2$$

$$\rightarrow w(s) = \frac{P(\frac{Ls^2}{2} - \frac{s^3}{6})}{EI}$$

Whiteboard notes

$$W(L) = q = \frac{PL^3}{3EI}$$

$$W(s) = \frac{P}{EI} \left(\frac{Ls^2}{2} - \frac{s^3}{6} \right)$$

$$W(s, t) = \frac{s^2 (3L - s)}{2L^3} q(t) \quad \leftarrow s^2 \left(\frac{L}{2} - \frac{s}{6} \right)$$

$$\psi(s)$$

$$\frac{L^3 P}{3EI} \frac{s^2 (3L - s)}{L^3 \cdot 2}$$

$$T = \frac{1}{2} \int_0^L \dot{w}^2(s, t) dm$$

$$\int_0^L \psi^2 \dot{q}^2 dm$$

Whiteboard notes

$$U = \frac{1}{2} \int \sigma \epsilon dV = \frac{1}{2} \int M d\theta$$

strengths
of
material

$$= \frac{1}{2} \int EI w''(s,t) \cdot w''(s,t) ds$$

$$\theta = w' \rightarrow d\theta \approx w''(s,t) ds$$

$$U = \frac{1}{2} \int EI w''^2 ds$$

3.6 Rayleigh's method

Beam model: effective mass

Use the static tip-load shape; normalize it at the tip.

$$w(s, t) = \psi(s)q(t), \quad \psi(s) = \frac{s^2(3L - s)}{2L^3}$$

$$m_{\text{eq}} = M + \int_0^L \rho A_b \psi^2(s) ds$$

$$k_{\text{eq}} = \int_0^L EI [\psi''(s)]^2 ds$$

Same coordinate and shape for both energies. Find ω_R .

3.7 Assumptions and result checks

Model and result checks

Coordinate ■ Units ■ Initial conditions

Whiteboard notes
